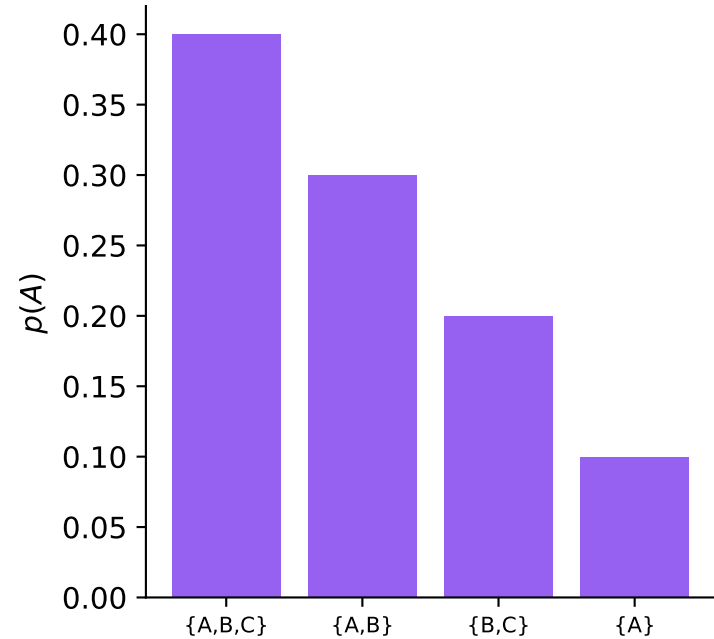
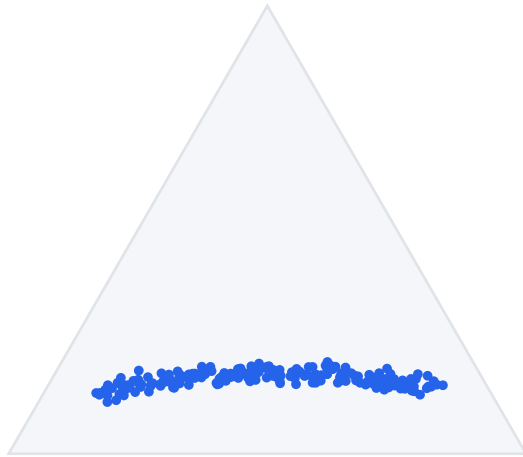


(1) draw a stratum $A \sim p(A)$
(from E1 occupancy)



(2) sample on F_A with
rank $k_A < |A| - 1$ (known tube)



(3) ground truth is known
(we designed k_A)

$$\{(x_j, A_j, k_j)\}$$

composition x_j on the simplex,
with its TRUE stratum A_j
and TRUE local dimension k_j

$\Rightarrow$ a non-manifold dataset with
known ground-truth dimension field